

Multimedia Course Overview



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MM Objectives



1. To provide an overview to the subject
2. To describe a number of fundamental concepts
3. To discuss the elements of multimedia systems
4. To explain the application of multimedia
5. To give a global structure of multimedia
 - Multimedia data representation
 - Multimedia data processing
 - Multimedia data compression
 - Multimedia data security
 - Multimedia data transmission
 - Multimedia mobile games
 - Multimedia human computer interaction

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What is Multimedia?

- **Multi-** means many; much; multiple
- **Medium means:**
 - An intervening substance through which something is transmitted or carried on
 - A means of mass communication such as newspaper, magazine, or television
- Multimedia is the combinations of text, graphic art, sound, animation, video and other kinds of elements.
- When a viewer of a multimedia presentation is allowed to control what elements are delivered
- and when, it is interactive multimedia.
- Multimedia is an inter-disciplinary subject because it involves a variety of different theories and skills:
 - these include computer technology, hardware and software;
 - arts and design, literature, presentation skills;
 - application domain knowledge.

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Overview of Multimedia Software Tools

- The categories of software tools briefly examined here are:

1. Music Sequencing and Notation

2. Digital Audio

3. Graphics and Image Editing

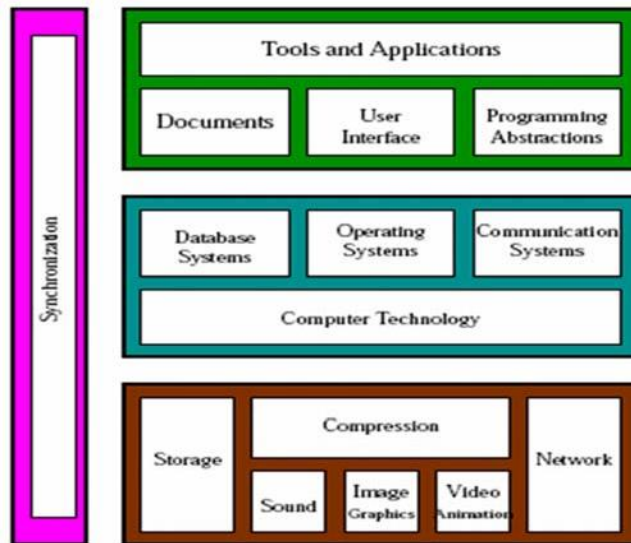
4. Video Editing

5. Animation

6. Multimedia Authoring

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Multimedia Global Structure



Application domain — provides functions to the user to develop and present multimedia projects. This includes *Software tools*, and multimedia projects *development methodology*.

System domain — including all supports for using the functions of the device domain, e.g., operating systems, communication systems (networking) and database systems.

Device domain — basic concepts and skill for processing various multimedia elements and for handling physical device.

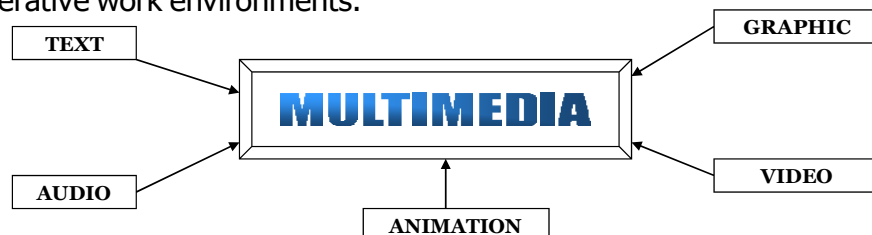
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Multimedia Global Structure

Multimedia and Computer Science:

– Graphics, HCI, visualization, computer vision, data compression, graph theory, networking, database systems. Multimedia and Hypermedia

- ❑ Multimedia involves multiple modalities of text, audio, images, drawings, animation, and video.
- ❑ Examples of how these modalities are put to use:
 - Video teleconferencing.
 - Distributed lectures for higher education.
 - Tele-medicine.
 - Co-operative work environments.



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What is a Multimedia System?

- **A system that involves:**
 - **generation** ➤ production/authoring tools
 - **representation** ➤ compression and formats
 - **storage** ➤ file system design
 - **transmission** ➤ networking issues
 - **search and retrieval** ➤ database management
 - **delivery** ➤ server design, streaming
- of multimedia
information**

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Multimedia Computing

- Multimedia systems involve some basic enabling techniques:
 - Multimedia data representation and compression.
 - Multimedia data processing and analysis.
 - Transmitting multimedia data through communication networks.
 - Multimedia database, indexing and retrieval.

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Challenges of Multimedia Computing

- Developing a successful multimedia system is non-trivial.
 - Continuous media types such as video need a lot of space to store and very high bandwidth to transmit.
 - They also have tight timing constraints.
 - Automatically analyzing, indexing and organizing information in audio, image and video is much harder than from text.
 - Multimedia involves many different research areas and needs more complex and more efficient algorithms and hardware platforms.

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Linear VS Non-Linear

LINEAR

- A Multimedia Project is identified as Linear when:
 - It is not interactive
 - User have no control over the content that is being showed to them.
- Example:
 - A movie
 - A non-interactive lecture / demo show

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Linear VS Non-Linear

NON-LINEAR

- A Multimedia Project is identified as Non-Linear when:
 - It is interactive
 - Users have control over the content that is being showed to them.
 - Users are given navigational control
- Example:
 - Games
 - Courseware
 - Interactive CD

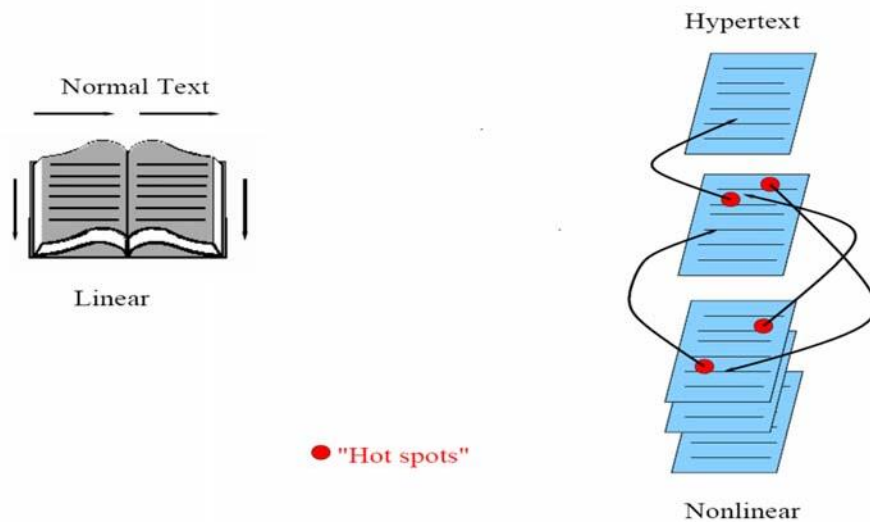
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Hypermedia and Multimedia

- A **hypertext** system: meant to be read nonlinearly, by following links that point to other parts of the document, or to other documents (Fig. 1.1)
- **HyperMedia**: not constrained to be text-based, can include other media, e.g., graphics, images, and especially the continuous media – sound and video.
 - The World Wide Web (WWW) — the best example of a hypermedia application.
- **Multimedia** means that computer information can be represented through audio, graphics, images, video, and animation in addition to traditional media.

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Multimedia Application

- Digital video editing and production systems.
- Electronic newspapers/magazines.
- World Wide Web.
- On-line reference works: e.g. encyclopaedia, games, etc.
- Home shopping.
- Interactive TV.
- Multimedia courseware.
- Video conferencing.
- Video-on-demand.
- Interactive movies.
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Importance of Multimedia

- There are a number of fields where multimedia could be of use. Examples are:-
 - Business
 - Education
 - Entertainment
 - Home
 - Public Places



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Importance of Multimedia

- **Business**
 - Use and Applications
 - Sales / Marketing Presentation
 - Trade show production
 - Staff Training Application
 - Company Kiosk



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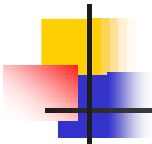
Importance of Multimedia

■ **Education**

- Use and Applications
 - Courseware / Simulations
 - E-Learning / Distance Learning
 - Information Searching



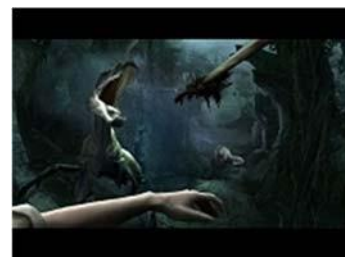
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Importance of Multimedia

■ **Entertainment**

- Use and Applications
 - Games (Leisure / Educational)
 - Movies
 - Video on Demand
 - Online



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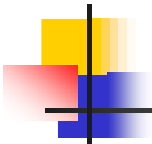
Importance of Multimedia

■ Home

- Use and Applications
 - Television
 - Satellite TV
 - SMS services (chats, voting, reality TV)



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Importance of Multimedia

■ Public Places

- Use and Applications
 - Information Kiosk
 - Smart Cards, Security



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Group Discussion



Multimedia System

A **multimedia system** refers to a combination of hardware and software designed to create, store, transmit, and present multimedia content. It allows for the **interactive use** of more than one type of media.

Properties of Multimedia Systems

1. Integration

- Combines different media types like text, graphics, audio, video, and animation.
- These components work together in a unified manner.

2. Interactivity

- Allows user control or influence over the content (e.g., clicking links, navigating a video).

3. Synchronization

- Media elements (e.g., audio with video) must be presented in a synchronized fashion.

4. Digital Representation

- All media types are represented in digital form for storage, manipulation, and transmission.

5. Temporality

- Some media are time-dependent (e.g., audio, video) and need to be played over time with specific timing constraints.

6. Hardware and Software Support

- Requires hardware such as cameras, microphones, and display devices.
- Also needs software like media players, codecs, and editing tools.

Characteristics of a Multimedia System

1. Media Integration

- Combines multiple types of media (text, graphics, audio, video, animation) into one system.
- Seamless integration ensures a rich user experience.

2. Interactivity

- Users can interact with the content (e.g., through buttons, navigation, input).
- Common in multimedia learning tools, games, and applications.

3. Temporal Synchronization

- Time-dependent media like audio and video must be synchronized (e.g., lip-sync in a video).
- Ensures smooth and coordinated playback.

4. Digital Representation

- All media components are stored and transmitted in digital form for consistency, editing, and delivery.
- Easier to manipulate and share across systems.

Components of a Multimedia System

1. Text

- The most basic form of multimedia.
- Used for titles, captions, labels, navigation, or as part of the content.
- Supports understanding and provides context.

2. Audio

- Includes music, narration, dialogue, and sound effects.
- Enhances engagement and supports emotional impact.
- Can be in formats like MP3, WAV, AAC.

3. Images/Graphics

- Still visuals like photographs, illustrations, or scanned drawings.
- Used for visual communication or decorative purposes.
- Common formats: JPEG, PNG, GIF.

4. Video

- Combines visuals and sound in motion.
- Used in tutorials, movies, demonstrations, and presentations.
- Common formats: MP4, AVI, MOV.

5. Animation

- A series of graphics or images displayed in sequence to simulate movement.
- Used in games, simulations, cartoons, educational content.
- Types: 2D, 3D animation, motion graphics.

System Components (Technical)

6. Hardware

- Physical equipment required to support multimedia:
 - Input Devices: Mouse, keyboard, camera, microphone.
 - Output Devices: Monitor, speakers, projector.
 - Storage Devices: Hard drives, SSDs, cloud storage.

7. Software

- Applications used to create, edit, store, and play multimedia:
 - Creation tools: Adobe Photoshop, Premiere Pro.
 - Playback tools: VLC Media Player, Windows Media Player.
 - Authoring tools: PowerPoint, Adobe Animate.

8. Multimedia Authoring Tools

- Used to combine different media types into a cohesive project.
- Allows for adding interactivity and navigation.
- Examples: Adobe Captivate, Unity, Flash, Director.

9. User Interface (UI)

- The part of the multimedia system that users interact with.
- Should be intuitive, responsive, and accessible.
- Includes buttons, menus, icons, and interactive elements.

10. Network and Connectivity

- Enables sharing, streaming, and collaborative multimedia applications.
- Includes the internet, LAN/WAN networks, and cloud platforms.
- Essential for web-based multimedia systems.

Areas Where Multimedia is Used

1. Education

- **E-learning platforms**, virtual classrooms, and digital study material.
- Visual aids like animations, videos, and simulations help explain complex concepts.
- Interactive quizzes and tutorials enhance learning experiences.

2. Entertainment

- Movies, TV shows, animation, video games, music streaming.
- Multimedia powers special effects, 3D animation, virtual concerts, and gaming environments.
- Essential for both traditional and digital entertainment industries.

3. Advertising & Marketing

- Interactive advertisements, video commercials, product demos.
- Digital marketing campaigns on social media and websites.
- Engages customers using appealing visuals and audio-visual storytelling.

4. Business & Corporate Sector

- Presentations, training modules, product launches, virtual meetings.
- Company websites and digital brochures use multimedia to promote services and improve communication.

5. Healthcare

- Medical simulations, 3D models for surgeries, telemedicine platforms.
- Patient education through animated videos and digital pamphlets.
- Used for diagnosis visualization (e.g., MRI and CT scan outputs).

6. Science & Technology

- Scientific visualization, research presentations, and technical animations.
- Used to model experiments, simulate environments, and present data.

7. Journalism & Mass Communication

- Multimedia news platforms combining text, images, video, and infographics.
- Enhances storytelling and real-time reporting on digital media.

8. E-commerce

- 3D product views, virtual try-on features, product videos.
- Enhances online shopping experience and increases customer trust.

9. Training & Simulation

- Used in defense, aviation, and industrial sectors for training through VR and simulations.
- Reduces risk and costs by simulating real-life scenarios.

10. Social Media & Communication

- Platforms like YouTube, TikTok, Instagram, and Facebook rely heavily on multimedia.
- Enables user-generated content, live streaming, and video messaging.

Future Scope

- **Augmented Reality (AR)** and **Virtual Reality (VR)** for immersive experiences.
- **Artificial Intelligence (AI)** in content creation and interaction.
- Growth in **interactive media**, **metaverse**, and **real-time streaming**.
- More advanced and engaging applications in **remote education**, **online services**, and **digital storytelling**.